

Executive Memo: Is the Reported 11% Recovery Improvement Real?

Collections Analytics — Data Analyst Assignment

What happened?

Recovery performance over the 7 full months of data (Jan–Jul 2026; August is a partial month and excluded from trend claims) is essentially flat, not improving. Naive month-on-month recovery swings noisily between –8.9% and +11.0% with no directional trend (Jan → Jul: –0.4% overall). After building a golden dataset that removes 500 exact-duplicate payment rows and nets out reversals in the month they occur, the pattern is unchanged in shape and slightly negative in level (Jan → Jul: –2.1%).

The reported "11% MoM improvement" matches exactly one month in the series — the Feb → Mar move (naive +11.0%, golden +10.0%) — generalized into a trend that the other six months don't support.

Where and which metrics moved (or didn't)

Metric	Jan '26	Jul '26	Trend
Contact rate (answered/outbound)	20.1%	19.4%	Flat
RPC rate	77.2%	77.7%	Flat
PTP rate	23.2%	21.6%	Flat
PTP kept rate	24.1%	24.6%	Flat
Conversion rate (paid/targeted)	41.4%	41.2%	Flat
Recovery / agent-hour	₹15,731	₹15,382	Flat
Net recovery (golden, monthly)	₹17.6 Cr	₹17.2 Cr	–2.1% over 7 mo

Every independently-defined metric — contact rate, RPC, PTP rate, PTP kept rate, population-normalized conversion rate, and recovery per agent-hour — is flat within a narrow band across all 7 months. This was cross-checked by vendor, calling hour, and attempt number: none show meaningful differentiation (contact rate range 19.2%–21.1% across all 15 vendors; flat across hour-of-day and attempt-number 1–7). Classification: Fact.

Why did it happen? (drivers)

- Portfolio mix (risk_segment, loan_type) at account-open level is stable month to month — no evidence of a materially different book being acquired. [Strong evidence]
- Targeted population size is stable (5,160–5,800 accounts/month) — the reported improvement is not an artifact of a shrinking denominator. [Fact]
- No channel, vendor, or agent shows a differential effect on outcomes in this data — meaning there is no current operational lever the data shows is working better than another. [Fact, on this dataset]
- Data-quality issues found (duplicate payments, reversal timing, disposition-code duplication, agent identity collisions) shift the absolute recovery level down ~7-9% but do not explain the shape of the (flat) trend. [Fact]
- We cannot rule out that the reporting layer generating the "11%" figure is simply comparing two arbitrary months rather than a rolling or seasonally-adjusted trend. [Hypothesis — recommend confirming with whoever produces that monthly report]

Is the reported 11% improvement real?

No. It is not a sustained trend by any of the 6+ independently-built definitions we tested (naive recovery, golden net recovery, conversion rate, contact rate, RPC, PTP rate/kept rate, recovery per agent-hour). The true Jan → Jul change is between –2% and flat, not +11% sustained.

Where should the ₹10 Cr go?

Recommended: AI Voice Automation, with a mandatory controlled pilot before full deployment.

Reasoning: contact rate (~20% of outbound calls connect) is the single largest funnel drop-off, and it is uniform across vendor, calling hour, and attempt number — i.e., the current human-calling operation has already converged to a ceiling that varying vendor/timing/attempts does not move. AI voice automation attacks this ceiling differently: it multiplies attempt throughput at a fraction of human agent-hour cost, expanding coverage of the ~80% currently unreachable population, while routing PTP negotiation and disputes to human agents. This is the option most likely to move a genuinely flat funnel, versus options (more agents, more telephony spend) that scale linear cost against a metric that isn't currently capacity-constrained by vendor or headcount.

- Expected incremental recovery: illustrative only — a 3–5 point lift in contact rate (from ~20% to 23-25%), applied to the ~80% currently-unreached targeted population, at the observed ~41% downstream conversion-to-payment rate, implies roughly ₹8–14 Cr/year incremental recovery. [Hypothesis — requires pilot data to confirm elasticity]
- Estimated cost: ₹10 Cr covers platform build/licensing + integration + a 90-day pilot across 2 vendor lines; assumes ~₹150–250 per 1,000 AI-voice minutes vs. ₹800–1,200 per 1,000 human agent-minutes (external assumption, not in this dataset).
- Expected ROI / breakeven: if incremental recovery lands at the low end (₹8 Cr/yr), breakeven is ~15 months; at the high end (₹14 Cr/yr), ~9 months. [Confidence: Low-Medium — wide range because this dataset shows no historical AI-voice channel to calibrate against]
- Key assumptions: (1) contact-rate ceiling is a dialing-capacity problem, not a borrower-willingness problem; (2) downstream PTP/kept/payment conversion for AI-originated contacts is comparable to human-originated contacts — untested; (3) no material increase in complaints/regulatory risk from AI voice — untested.
- Downside scenario: if AI-originated contacts convert to PTP/payment at materially lower rates than human contacts (plausible if borrowers are less responsive to automated voice), the realistic floor is closer to ₹2-3 Cr/yr incremental recovery, pushing breakeven past 3 years — at which point the better use of the ₹10 Cr would have been more agents or targeting.

Given the wide uncertainty band above, we recommend deploying ₹1.5–2 Cr of the ₹10 Cr as a 90-day controlled pilot (treatment vs. matched control accounts) before committing the remainder — see the Counterfactual/Experimentation design in the full analysis notebook.

Statistical checks & counterfactual (full detail in notebook)

- Simpson's paradox: tested directly by risk_segment and loan_type — no subgroup shows a trend that the aggregate could be masking. [Fact]
- Mix/cohort/selection/survivorship effects: portfolio mix, targeted-population DPD, and population size are all stable month to month — none of these explain the flat trend. [Fact]
- Attribution-window bias: the schema has no ground-truth link between a specific contact and a resulting payment; any channel attribution (including in this memo) is last-touch and should be read as directional, not precise. [Hypothesis — genuine blind spot]
- Counterfactual (targeting-strategy change): we tested for an actual strategy break in the data and found none — strategy_version and target_definition mix are flat throughout. A Difference-in-Differences design is documented for when leadership can supply an actual change date; fabricating a break point the data doesn't support would be worse than flagging the gap.

Confidence & what would change our mind

High confidence: the reported 11% is not a sustained trend (multiple independent metrics agree, and Simpson's/mix/selection/survivorship effects are all ruled out). Low-to-medium confidence: the ₹10 Cr recommendation, because this dataset contains no existing AI-voice channel to calibrate elasticity — the pilot is not optional, it is the load-bearing part of this recommendation.